

14-01-26
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Module-1

Descriptive Statistics.

Descriptive statistics:

- Statistics is the discipline concerned with collection, organisation and analysis - interpretation of data.
- In modern computing application such as data analysis, machine learning, business intelligence and research, large quantity/amount of data must be summarized efficiently before advanced analysis.

Diagrammatic representation of data

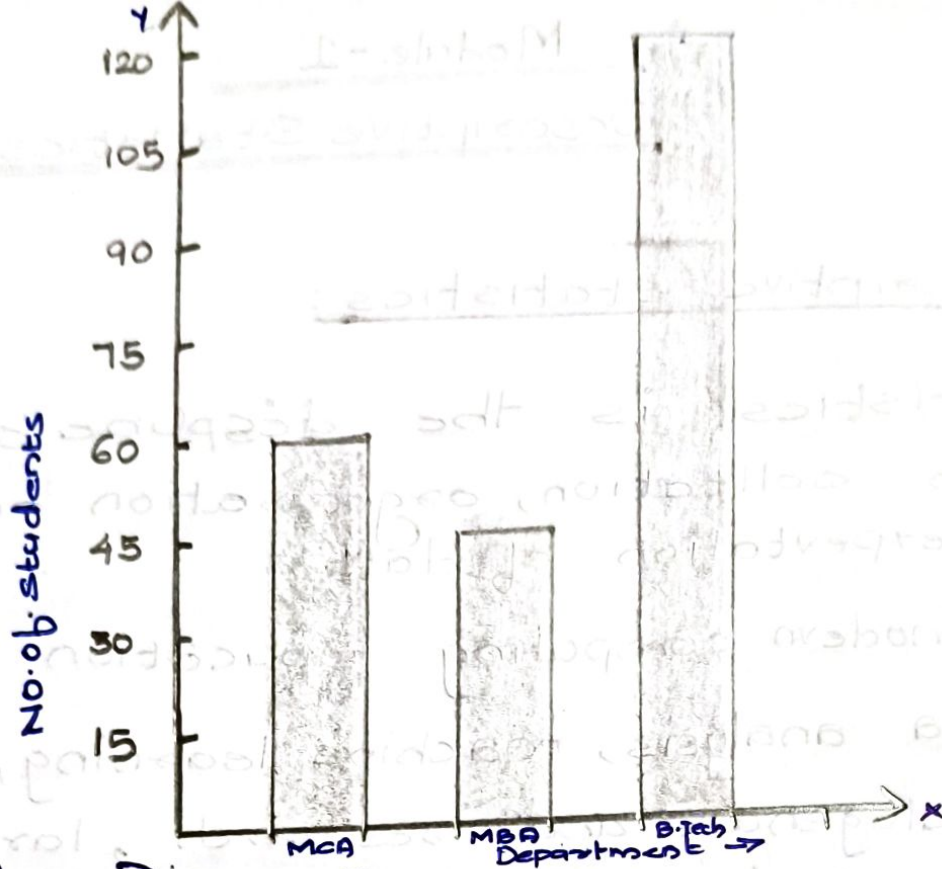
Diagrammatic methods present data in a simple and attractive manner, suitable for comparison.

Bar Diagram

It is used for discrete or categorical data.

Example 1 : Number of students in department

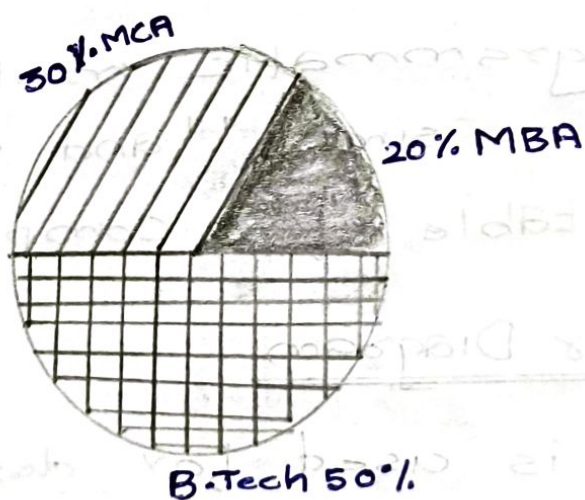
Department	Student
MCA	60
MBA	45
B.Tech	120



Pie Diagram

It represents data as a part of whole.

Ex: Example 2: If MCA students are 30%, MBA 20%, B.Tech 50%.



Each sector angle = percentage $\times 60^\circ$

Graphical Representation of Data

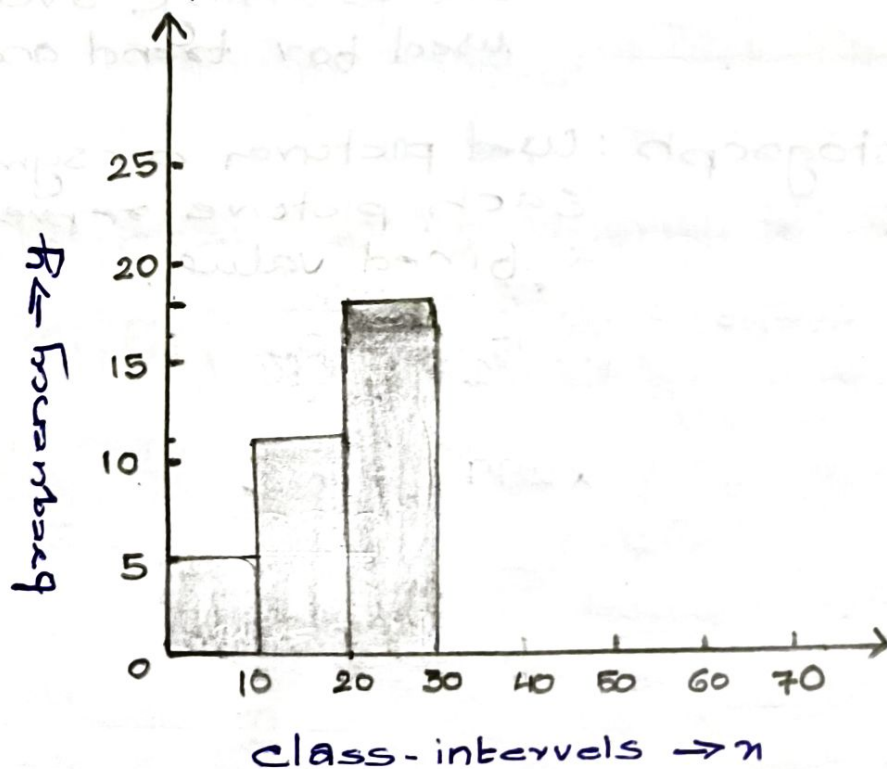
Graphs are more suitable for continuous data.

1) Histogram

Is a visual representation of the distribution of quantitative data or uses bars to show the distribution of continuous numeric data.

Example 13 :

class interval	frequency
0-10	5
10-20	12
20-30	18



Identify different type of diagrams and graphs.

eg: Histogram, Piechart, barplot

To represent data visually, we use different diagrams and graphs. Each type is suitable for a specific kind of data and purpose.

1. Bar diagram: Uses rectangular bars. Bars can be vertical or horizontal, used to compare quantities.
2. Pie Diagram: A circular chart divided into sectors. Each sector shows a % or proportion.
3. Line Diagram: Data points joined by lines shows change over time, used for trend analysis.
4. Pictograph: Uses pictures or symbols. Each picture represents a fixed value.

Measures of central tendency

It represent the central value of a data set.

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Arithmetic mean

Most commonly used measure of central tendency. Arithmetic mean of a set of observations is, the sum divided by the total number of observations

Eg: The Arithmetic mean ' $\bar{x}$ ' of ' n ' observation $x_1, x_2, \dots, x_n$ is given by:

$$\begin{aligned}\bar{x} &= \frac{1}{n} (x_1 + x_2 + \dots + x_n) \\ &= \frac{1}{n} \sum_{j=1}^n x_j\end{aligned}$$

In case of frequency observation $x_i | f_i$, $i = 1, 2, \dots, n$ where ' f_i ' is the frequency of the variable ' x_i ' then

$$\bar{x} = \frac{x_1 f_1 + x_2 f_2 + \dots + x_n f_n}{f_1 + f_2 + \dots + f_n}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i f_i}{\sum_{i=1}^n f_i}$$

In case of grouped or continuous frequency distribution, x is taken as the middle value

of the corresponding class.

Q Find the arithmetic mean of the frequency distribution:

x	1	2	3	4	5	6	7
f	5	9	12	17	14	10	6

Ans

x_i	f_i	$x_i f_i$
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
Σ	$N=73$	299

$\bar{x}$; Arithmetic mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n x_i f_i$$

$$= \frac{1}{73} \times 299 =$$

$$= \underline{\underline{409}}$$

Q Calculate the A.M marks from the following tables.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No: of Students	12	18	27	20	17	6

Ans:

class	No: of students f_i	x_i	$x_i f_i$
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765
50-60	6	55	330
Σ	100		2800

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n x_i f_i$$

$$= \frac{1}{100} \times 2800$$

$$= \underline{\underline{28}}$$

In case of grouped or continuous frequency distribution, the AM can be calculated by taking

$$d_i = \frac{x_i - A}{h}$$

Where 'A' is an arbitrary point (usually the value of x corresponding to the middle part of distribution).

and h is the common magnitude of class interval. Then

$$\bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i$$

Q3 Calculate the mean for the following distribution:

Class interval	Frequency
0-8	8
8-16	7
16-24	16
24-32	24
32-40	15
40-48	7

Ans:

Class interval	midvalue	Frequency	$d_i = \frac{x_i - A}{h}$	$f_i d_i$
0-8	4	8	-3	-24
8-16	12	7	-2	-14
16-24	20	16	-1	-16
24-32	28	24	0	0
32-40	36	15	1	15
40-48	44	7	2	14

let $A = 28$

$$\bar{x} = 28 + \frac{8}{77} \times -25$$

$$= \underline{\underline{25.4025974}}$$

Mean of Composite series.

If $\bar{x}_1, \bar{x}_2$ are means of 2 component series of size n_1 and n_2 , then the mean $\bar{x}$ of the composite series on obtained on combining the component series is given by the formula:

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Q4 The average salary male employees in a firm was RS 520 and that of females was RS 420, The mean salary of all employees was RS 500. Find the % of male and female employees.

Ans: Given:

Average salary of male $\bar{x}_1 = \text{RS } 520$

Average salary of female $\bar{x}_2 = \text{RS } 420$

Average salary of All employees $\bar{x} = \text{RS } 500$

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$500 = \frac{n_1 \times 520 + n_2 \times 420}{n_1 + n_2}$$

$$500(n_1 + n_2) = n_1 520 + n_2 420$$

$$500n_1 + 500n_2 = n_1 520 + n_2 420$$

$$20n_1 = 80n_2$$

$$n_1 = 4n_2$$

$$\frac{n_1}{n_2} = 4$$

$$\Rightarrow n_1 = 4n_2$$

$$\therefore \text{Total employee} = n_1 + n_2$$

$$= 4n_2 + n_2$$

$$= 5n_2$$

$$\% \text{ of female} = \frac{\text{no. of female}(n_2)}{\text{Total employee}} \times 100$$

$$= \frac{n_2}{5n_2} \times 100$$

$$= \underline{\underline{20\%}}$$

$$\% \text{ of male} = \underline{\underline{80\%}} \quad [100 - 20]$$

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Median

Median of a distribution is the value of the variable which divides it into 2 equal parts. It is the value which exceeds and exceeded by the same no. of observation.

In case ungrouped data if no. of observation is odd then median is the middle value after the values arranged in ascending or descending order of magnitude.

In case of even no. of observation there are two middle terms and the median is taken by taking the arithmetic mean of the middle values

Eg:

1) 25, 20, 15, 35, 18

• First arrange the given observation in ascending or descending order:

→ 15, 18, 20, 25, 35

Total number of observation $N = 5$

Since 'N' is odd, so take the middle value.

Here, median = 20

2) 8, 20, 50, 25, 15, 30

• Arrange in descending order

50, 30, 25, 20, 15, 8

Here $N = \underline{6}$, since it is even

Median = Arithmetic mean of middle terms

$$= \frac{25+20}{2}$$

$$= \underline{\underline{22.5}}$$

In case of discrete frequency distribution median is obtained by considering cumulative frequencies.

The steps for calculating median are given below:

Step 1: Find $\frac{N}{2}$, where 'N' is $\sum f_i$

Step 2: See the [less-than] cumulative frequency just greater than $\frac{N}{2}$

Step 3: The corresponding value of x is the median

Q1. Obtain the median for the following frequency distribution

x :	1	2	3	4	5	6	7	8	9
f :	8	10	11	16	20	25	15	9	6

n	Fi	CF
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	80
7	15	105
8	9	114
9	6	120

$$N = 120$$

$$\frac{N}{2} = 60$$

$$\text{median} = 5$$

In the case of continuous frequency distribution, the class corresponding to the cumulative frequency just greater than $N/2$ is called the median class, the value of median is obtained by following formula:

$$\text{Median} = l + \frac{h}{f} \left[\frac{N}{2} - C \right]$$

where l is the lower limit of the median class

f is the frequency of the median class

h is the magnitude of the median class

C is the CF of the class preceding the median class

$$N = \sum F$$

Q2.

Find the median wage of the following distribution:

Wage (in Rs): 20-30 30-40 40-50 50-60 60-70

No. of labours: 3 5 20 10 5

soln

Class (Wage)	Frequency (No. of labours)	CF
20-30	3	3
30-40	5	8
40-50	20	28
50-60	10	38
60-70	5	43

$$N = 43$$

$$\frac{N}{2} = 21.5$$

∴ Median class : 40-50

$$l = 40, h = \text{upper limit} - \text{lower limit} = 20 \quad c = 8$$

$$\text{Median} = l + \frac{h}{f} \left[\frac{N}{2} - c \right]$$

$$= 40 + \frac{10}{20} \times \left[\frac{43}{2} - 8 \right]$$

$$= 40 + \frac{10}{20} \times \frac{27}{2}$$

Mode

Mode is the value that occurs most frequently in a set of observations.

Eg: In the following frequency distribution

x	1	2	3	4	5	6	7	8
f	4	9	16	25	22	15	7	3

Here Mode = 4 as it occurred 25

In case of continuous frequency distribution mode is given by:

$$\text{Mode} = l + \frac{h(f_0 - f_0)}{(f_0 - f_0) - (f_2 - f_1)} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

Where: • l is the lower limit

• h is the magnitude

• f_1 the frequency of modal class.

• f_0 and f_2 are the frequency of the classes preceding and succeeding the modal class respectively

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Q3

In a factory employing 3,000 persons, 5 per cent earn less than RS. 3 per hour, 580 earn from RS 3.01 to RS. 4.50 per hour, 30 percent earn from RS 4.51 to RS 6.00 per hour, 500 earn from RS. 6.01 to RS 7.50 per hour, 20 percent earn from RS 7.51 to RS. 9.00 per hour, and the rest earn Rs 9.

or more per hour. What is the median wage?

Class (wage)	Frequency f_i	Cf
Below 3	150	150
3.01 - 4.50	580	730
4.51 - 6.00	900	1630
6.01 - 7.50	500	2130
7.51 - 9.00	600	2730
9.01 & above	270	3000

$$N = 3000$$

$$\frac{N}{2} = 1500$$

$$\therefore \text{Median class} = 4.51 - 6.00$$

$$l = 4.51$$

$$h = 6.00 - 4.51$$

$$f = 900$$

$$c = 730$$

$$\text{Median} = l + \frac{h}{f} \left[\frac{N}{2} - c \right]$$

$$= 4.51 + \frac{1.49}{900} \left[\frac{3000}{2} - 730 \right]$$

$$\begin{aligned}
 &= 4.51 + \frac{1.49}{900} [1500 - 730] \\
 &= 4.51 + \frac{1.49}{900} \times 770 \\
 &= 4.51 + \frac{1.49}{90} \times 77 \\
 &= 4.51 + 1.274 \\
 &= \underline{\underline{5.78}}
 \end{aligned}$$

Q4. An incomplete frequency distribution is given as follows.

Variable	Frequency
10-20	12
20-30	30
30-40	?
40-50	65
50-60	?
60-70	25
70-80	18

Given that the median value is 46 determine the missing frequency using the median formula.

Ans:

$$\text{Median value} = 46$$

$$\text{Median class} = 40-50$$

$$\text{median} = l + \frac{\left(\frac{N}{2} - c\right) \times h}{f}$$

$$N = 229$$

$$\frac{N}{2} = 114.5$$

$$l = 40$$

$$h = 10$$

$$f = 65$$

$$c = 12 + 30 + f_1$$

$$= 42 + f_1$$

$$\text{median} = 46$$

$\Rightarrow$

$$46 = 40 + \frac{114.5 - (42 + f_1)}{65} \times 10$$

$$6 = \frac{(72.5 - f_1)}{65} \times 10$$

$$\frac{6 \times 65}{10} = 72.5 - f_1$$

$$39 = 72.5 - f_1$$

$$f_1 = 72.5 - 39$$

$$= \underline{\underline{33.5}}$$

EF:

$$12 + 30 + 33.5 + 65 + f_2 + 25 + 18 = 229$$

$$183.5 + f_2 = 229$$

$$f_2 = 229 - 183.5 = 45.5$$